

Answer: D

- 6 The skateboarder will experience the maximum force by the half-pipe at the most bottom point, as the speed will be maximum at that point. Let the radius of the half-pipe be r .

By conservation of energy,

Gain in kinetic energy = loss in gravitational potential energy

$$\frac{1}{2}mv^2 - 0 = mgr$$

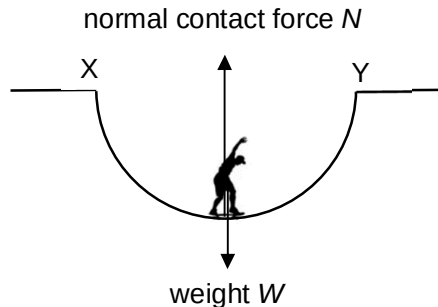
$$v^2 = 2gr$$

The normal contact force and the gravitational force on the skateboarder contributes to the centripetal force acting on him.

$$F_c = N - W$$

$$\begin{aligned} N &= \frac{mv^2}{r} + mg \\ &= m \left(\frac{2gr}{r} + g \right) \\ &= 3mg \end{aligned}$$

Answer: C



- 7 The gravitational field strength at a point at a distance r away from the point mass